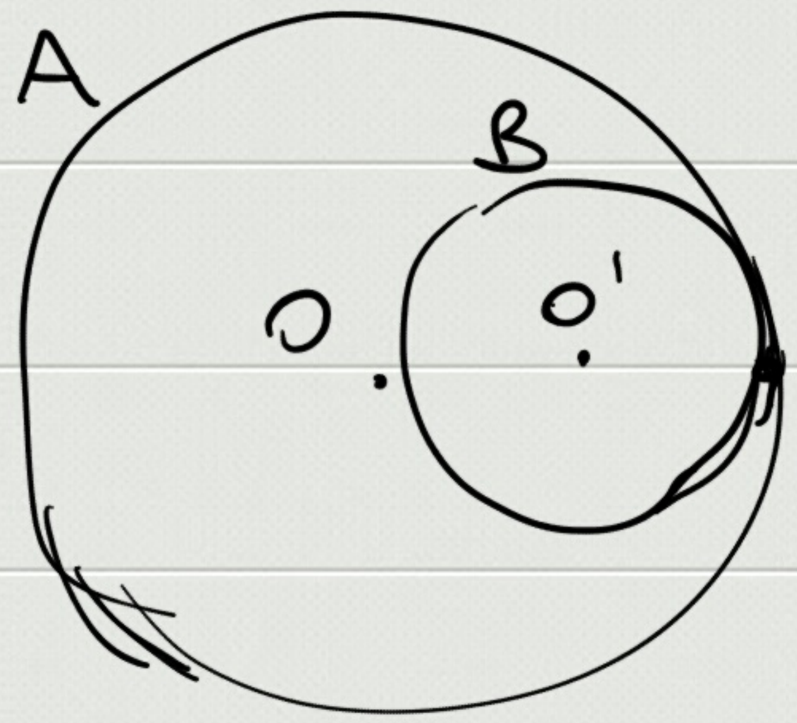
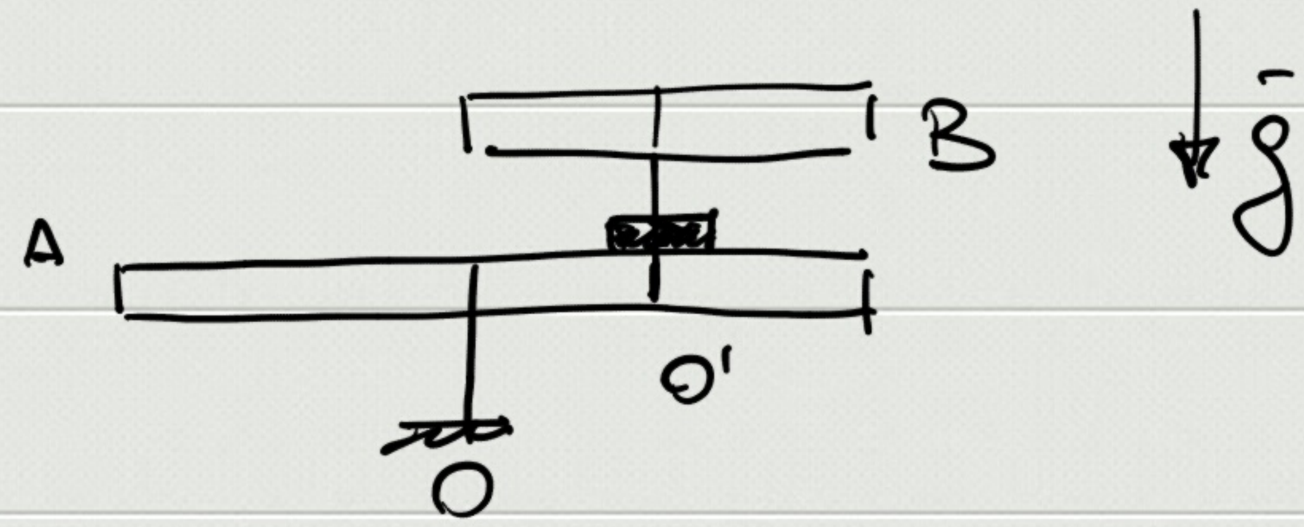
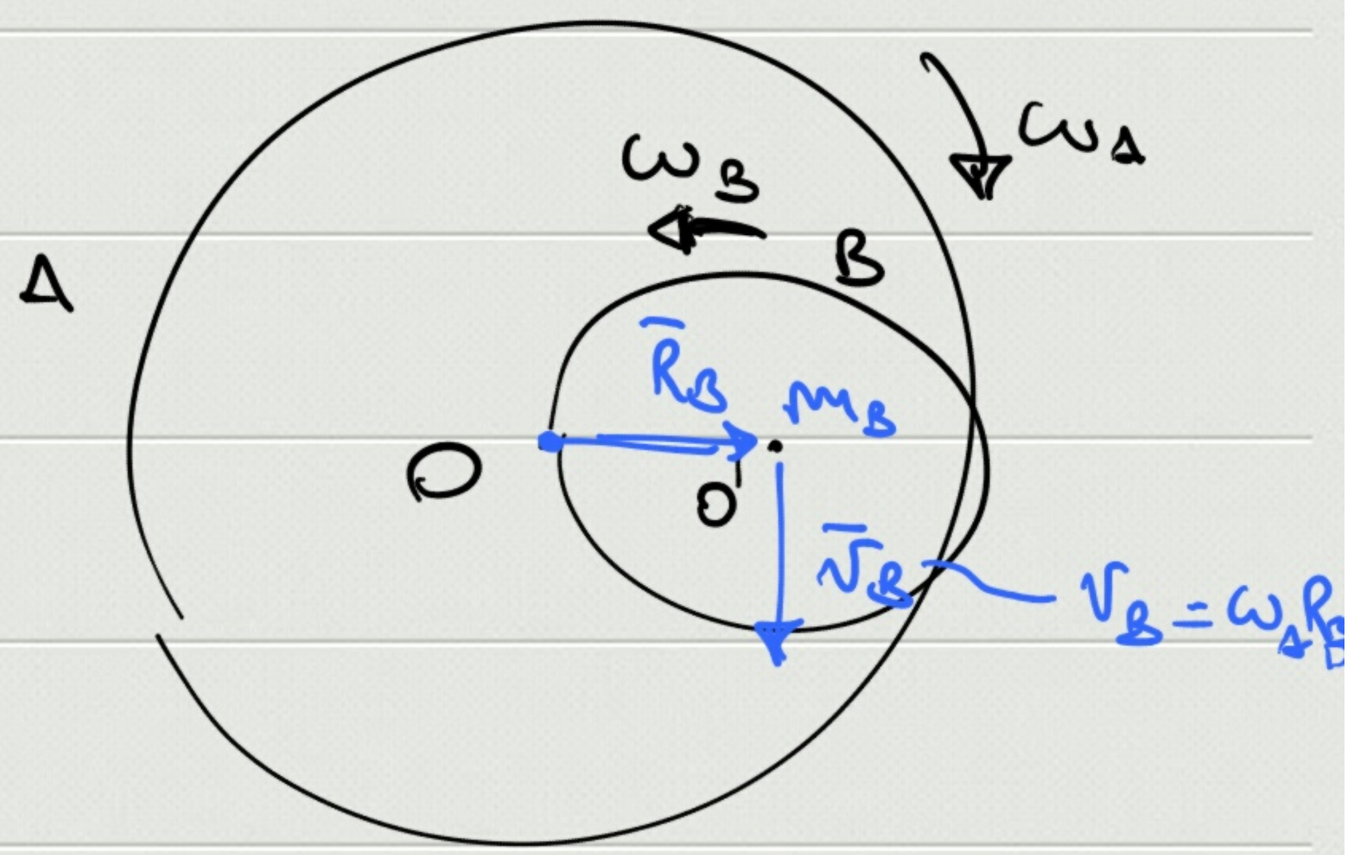




$m_A = 20\ kg \qquad m_B = \frac{m_A}{4} \qquad R_A = 1\ m \qquad R_B = \frac{R_A}{2}$

$\omega_{0A} = \omega_{0B} = 0 \qquad M_{O'} = 10\ Nm$

$\omega_B = ? \qquad \omega_A = 5\ rad/s$



$\bar{L} = const = 0$

$\bar{L}_O = 0 = I_A \bar{\omega}_A + I_B \bar{\omega}_B$

$= I_A \bar{\omega}_A + I_B \bar{\omega}_B + \bar{R}_B \times m_B \bar{v}_B \quad * \left(\bar{L}_O = \bar{L}'_{CM} + \bar{L}_{O,CM}\right)$

$= I_A \omega_A - I_B \omega_B + R_B m_B v_B \quad *$

$= I_A \omega_A + I_B \omega_B + R_B m_B v_B$ <u>Nb</u>

$$\Rightarrow \quad 0 = \frac{1}{2} m_A R_A^2 \omega_A - \frac{1}{2} m_B R_B^2 \omega_B + R_B m_B \omega_A R_B$$

$$\Rightarrow \omega_B = +18\,\omega_A = 90 \text{ rad/s}$$

$\alpha_A = ?$

$\frac{d}{dt}$

$\Rightarrow \alpha_A = 0.\bar{8} \text{ rad/s}^2$

- $\cancel{\alpha_A = 0}$

- $I_A \alpha_A - I_B \alpha_B + R_B m_B R_B \alpha_A = 0$ $\qquad \alpha_B = \frac{M_{O'}}{I_B}$ *

- $\underbrace{\alpha_A = \frac{1}{18} \alpha_B}$ $\qquad \alpha_B = \frac{M_{O'}}{I_B}$ * $\qquad M_{O'} = I_B \alpha_B$

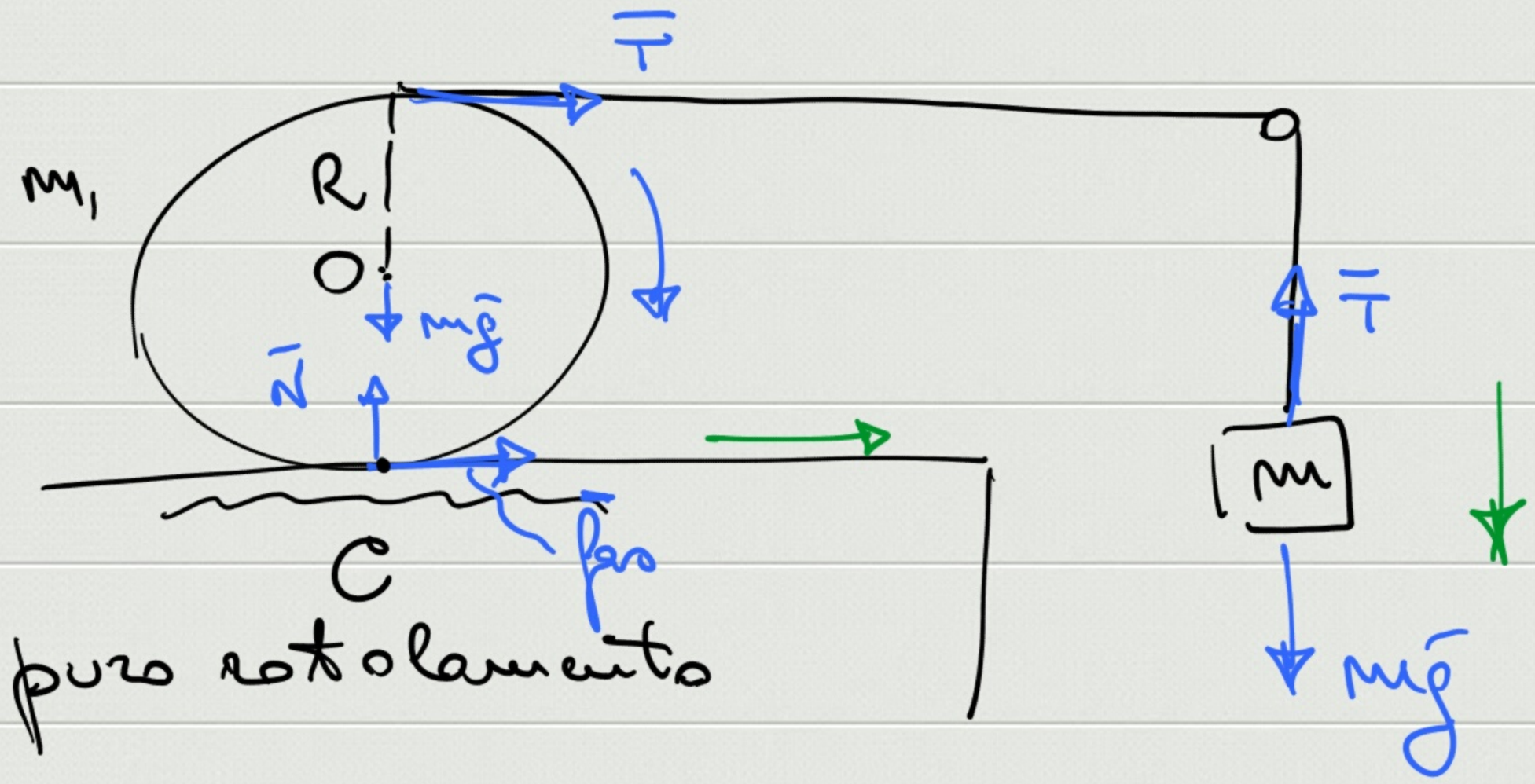
- $\cancel{\alpha_A = \frac{M_{O'}}{I_A + (I_B + m_B R_B^2)}}$

$\omega_A^* \; (\omega_B^* = 0) = ?$

$\boxed{L_O = 0}$

$\omega_A^* = \omega_A$ $\qquad \omega_A^* = -\omega_A$

$\boxed{\omega_A^* = 0}$ $\qquad \omega_A^* = -\omega_B$



$a_{CM} = ?$

$$\bar{M}^{E}_{polo} = I_z \bar{\alpha}$$

$$\cancel{-RT + Rf_{as} = \frac{1}{2} m R^2 \alpha}$$

$$2RT = \left(\frac{1}{2} m R^2 + m R^2\right)\alpha \quad *$$

$$\cancel{2RT = \frac{1}{2} m R^2 \alpha}$$

$$RT - Rf_{as} = \frac{1}{2} m R^2 \alpha \quad *$$

$$\begin{cases} mg - T = ma \\ T + f_{as} = m a_{CM} \\ \cancel{R}T - \cancel{R}f_{as} = \frac{1}{2} m R^2 \alpha = \frac{1}{2} m R^{\cancel{2}} \frac{a_{CM}}{\cancel{R}} \\ a = a_T = 2 a_{CM} \end{cases}$$

$$\begin{cases} mg - T = 2ma_{CM} \\ 2T = \frac{3}{2} m a_{CM} \end{cases} \Rightarrow T = \frac{3}{4} m a_{CM}$$

$$\cancel{m}g - \frac{3}{4}\cancel{m}a_{CM} = 2\cancel{m}a_{CM} \Rightarrow \boxed{a_{CM} = \frac{4}{11} g}$$

$\mu_s = ?$

$$f_{as} = m a_{CM} - T = \frac{1}{4} m a_{CM} = \frac{1}{11}\cancel{m}g \leq \mu_s \cancel{m}g$$

$$\Rightarrow \boxed{\mu_s \geq \frac{1}{11}}$$